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Development of sustainable CO_2 conversion processes for the methanol production

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Abstract

Utilization of CO_2 feedstock through CO_2 conversion for producing valuable chemicals as an alternative to sequestration of the captured CO_2 is attracting increasing attention in recent studies. Indeed, the methanol production process via thermochemical CO_2 conversion reactions is considered a prime candidate for commercialization. The aim of this study is to examine two different options for a sustainable methanol plant employing the combined reforming and CO_2 hydrogenation reactions, respectively. In addition, process improvement strategies for the implementation of the developed processes are also considered. The two methanol plants are developed using Aspen Plus®, the commercial process simulator. The net CO_2 flows and methanol production costs are evaluated using ECON® and compared with those of the conventional methanol plant, which uses two-stage reforming. It is verified that the combined reforming process has to be integrated with the existing conventional methanol plant to obtain a reduced CO_2 emission as well as lowered production costs. On the other hand, the CO_2 hydrogenation based methanol plant could achieve a reduction of net CO_2 emission at a reasonable production cost only with utilization of renewable energy resources (hydroelectric power and biomass) for the H_2 feedstock.

Keywords: CO_2 conversion, Methanol production, Sustainable process design, Process integration, Renewable energy utilization

1. Introduction

As CCUS (Carbon capture, utilization and storage) is becoming a popular R&D theme due to the mounting concerns about climate change, CO_2 conversion has attracted much attention lately. Basically, the aim of CO_2 conversion is to utilize the concentrated CO_2 (from CO_2 capture) as a feedstock to produce valuable chemicals via various conversion processes including thermochemical reaction, electrochemical reduction, photo-catalytic reduction, and bio-enzymatic reduction (Adebajo and Frost, 2012; Taheri Najafabadi, 2013). However, for CO_2 conversion to become widely practical, significant hurdles will need to be overcome. Slow kinetics and high energy demand associated with most CO_2 conversion reactions can lead to high economic penalties and additional CO_2 emission.

In spite of much research done on CO_2 conversion reactions in recent years, works on their process development, optimization, and analysis of their economic feasibility and impact on reduction have been scarce. In this paper, two thermochemical CO_2 conversion reactions, namely the combined reforming and CO_2 hydrogenation, are investigated. Designs of sustainable methanol plants employing these reactions and the conventional methanol plant are developed, respectively. Moreover, the net CO_2 flow and economic feasibility, crucial factors for sustainable process development, of the two methanol plants are compared. Finally, for those infeasible cases, either in terms of net CO_2 flow or economics, strategies are suggested to make them feasible.

2. Process description

In this study, three methanol production processes are developed and compared with one another. Two of the three utilize CO_2 as feedstock and the third represents a conventional process for methanol production. In the conventional process, syngas (the mixture of H_2 and CO) is produced through two-stage reforming (steam reforming and partial oxidation) and methanol is synthesized from the produced syngas (Dahl et al., 2014). The combined reforming (also referred to as mixed reforming or bi-reforming) can also produce syngas for methanol synthesis as in the conventional case, but it uses CO_2 as well as methane as feed and there is only one reforming stage combining the steam reforming and dry reforming steps (Holm-Larsen, 2001; Olah et al., 2009). Finally, methanol can be synthesized from CO_2 hydrogenation using CO_2 and H_2 (Van-Dal and Bouallou, 2013). The reactions taking place in each of the three processes are listed in Table 1.

The two CO_2 conversion reactions (see Table 1) are based on mature technologies already tested in large scale and therefore have the potential for further development. Most other CO_2 conversion options are currently in small-scale experimental stage. For example, a commercial-scale methanol plant (3,030 MTPD) adopting combined reforming has started up in Iran in 2004 (Aasberg-Petersen et al., 2008) and another commercial methanol plant (5 million liters per year) adopting CO_2 hydrogenation has been successfully launched in Iceland in 2011 (CRI, 2011).

Flowsheet simulations of the three methanol plant have been developed in Aspen plus® with the design specifications taken from Holm-Larsen (2001), Van-Dal and Bouallou (2013), and Dahl et al. (2014), respectively. The feed and reactor (including the primary reformers and the CO_2 hydrogenation reactor) operating conditions are optimized further through the simulator (see Table 2).

Table 1. Three different reactions for the methanol production

	Reaction equation	Operating conditions
Combined reforming	$\mathrm{CH_4} + H_2O \rightarrow 3H_2 + CO, \Delta\mathrm{H}^{\circ} = 206.3\ \mathrm{kJ/mol}$	850~950 °C
	$\mathrm{CH_4} + CO_2 \rightarrow 2H_2 + 2CO, \Delta\mathrm{H}^{\circ} = 247.3\ \mathrm{kJ/mol}$	20~30 bar
CO_2 hydrogenation	$\mathrm{CO_2} + 3H_2 \rightarrow CH_3OH + H_2O$	200~300 °C
	$\Delta\mathrm{H}^{\circ} = -49.32\ \mathrm{kJ/mol}$	50~100 bar
Two-stage reforming	$\mathrm{CH_4} + H_2O \rightarrow 3H_2 + CO, \Delta\mathrm{H}^{\circ} = 206.3\ \mathrm{kJ/mol}$	850~950 °C
	$\mathrm{CH_4} + 0.5O_2 \rightarrow H_2 + CO, \Delta\mathrm{H}^{\circ} = -35.6\ \mathrm{kJ/mol}$	20~30 bar

Table 2. Feed and operating conditions of three methanol plants

	S/C_nH_m	CO_2/C_nH_m	O_2/C_nH_m	H_2/CO_2	T (°C)	P (bar)
Combined reforming	1.79	0.29	-	-	950	22.5
CO_2 hydrogenation	-	-	-	3.00	200	70
Two-stage reforming	1.80	-	0.46	-	890	25

3. Process evaluation and results

In principle, net CO_2 flow and economic feasibility are essential criteria for discussing the feasibility of CO_2 conversion processes.

$$Net\ CO_2\ flow = \sum_i F_{CO_2,i}^{Generated} - F_{CO_2}^{Input} \quad (1)$$

The feasible CO_2 conversion process must achieve a reduction of net CO_2 flow defined as in Eq.(1). Here, the input flow ($F_{CO_2}^{Input}$) is being converted while the generated CO_2 ($F_{CO_2,i}^{Generated}$) is the amount emitted through purge (direct emission) and/or produced for energy used by the process (indirect emission) (Frauzem et al., 2015). Also, positive profit should be guaranteed, based on the mass and energy balance information of the process simulation. The cost of each methanol plant is evaluated by using ECON®. For the costs and energy supply, the equivalent of CO_2 generated is calculated. For the CO_2 feedstock, it is assumed to be captured at a H_2 plant (Metz et al., 2005). H_2 is produced via steam methane reforming (SMR) (Bartels et al., 2010), and the power demand is supplied from the conventional gas-fired power plant (U.S. EIA, 2013). Also, the natural gas is assumed to be 3.915 USD/MMBTU.

Table 3 lists the net CO_2 flow and the corresponding methanol production costs. However, its net CO_2 flow is positive because of the high heat demand in the reformer that resulted in significant CO_2 emission from the reformer furnace (accounting for 92.8 % of total emission, see Table 4). The CO_2 hydrogenation based methanol plant generates the most CO_2 because of the large demand for H_2 which is very expensive and gives significant indirect emission of CO_2 (97.8 %, see Table 4). Therefore, both of the CO_2 conversion based methanol plants are not feasible as is, and follow-up strategies are required.

Table 3. Net CO_2 flow and the methanol production cost of the three methanol plants

	Net CO_2 flow (kg_{CO2}/t_{MeOH})	Production cost ($\$/t_{MeOH}$)
Combined reforming	253.56	227.68
CO_2 hydrogenation	1,202.07	596.72
Two-stage reforming	311.36	260.31

Table 4. CO_2 flows of three methanol plants

	CO_2 input rate (kg_{CO2}/t_{MeOH})	Direct CO_2 emission		Indirect CO_2 emission	
		Rate (kg_{CO2}/t_{MeOH})	Portion (%)	Rate (kg_{CO2}/t_{MeOH})	Portion (%)
Combined reforming	375	583	92.8	45	7.2
CO_2 hydrogenation	1,456	60	2.2	2,599	97.8
Two-stage reforming	0	264	84.9	47	15.1

4. Strategies toward the sustainable CO_2 conversion process

A new standalone installation of neither of the developed methanol plants utilizing CO_2 seems promising owing to their positive net CO_2 flow and/or too high production costs. To make them more attractive, process integration and utilization of renewable energy sources are considered.

4.1. Process integration of the combined reforming-based methanol plant with an existing methanol plant

Although an economic advantage is seen from the combined reforming based methanol plant, the problem is that more CO_2 is released from the process than the input amount. Instead of an independent implementation, if integration of two methanol plants in operation, a conventional methanol plant and the CO_2 conversion process, is considered, actual reduction of CO_2 emission can be achieved. In fact, both the conventional methanol plant and the combined reforming based methanol plant produce the syngas necessary for synthesizing methanol. Therefore, for given methanol production capacity, the syngas production rate from the two-stage reforming processing section can be reduced and the reduced amount can be supplemented by the combined reforming section. This “process integration” strategy is illustrated in Figure 1.

For various integration ratios (0.1~0.7, ratio 0.7 means 70 % of original syngas production rate at the two-stage reforming process is generated by combined reforming process and returned to the methanol synthesis section), several performance indices including the CO_2 abatement rate, CO_2 credit benefit, and the methanol production cost saving have been calculated and given in Table 5. Here, 4.63 EUR/t_{CO2} of carbon credit is applied in the calculation of the carbon emission credit benefit.

Consequently, increasing the process integration ratio (producing more syngas by the combined reforming process) leads to a decrease in CO_2 emission and an increase in the production cost saving. The reason is that syngas production via combined reforming is less costly and also emits less CO_2 than the two-stage reforming.

Eventually, depending on the desired CO_2 reduction rate, the proper integration ratio can be decided. This way one can obtain a more sustainable solution without any economic sacrifice. In addition, extra income may be earned from the carbon emission credit trading but it is rather minor (0.166 USD/t_{MeOH} for 50 % integration) at the current rate of carbon emission credit.

Table 5. CO_2 and economic-related performances with respect to the integration ratio

Integration ratio	0.1	0.3	0.5	0.7
CO_2 abatement rate (%)	1.82	5.45	9.09	12.7
Carbon credit benefit $/tMeOH	0.0332	0.0997	0.166	0.233
Production cost saving $/tMeOH	0.765	2.30	3.83	5.36

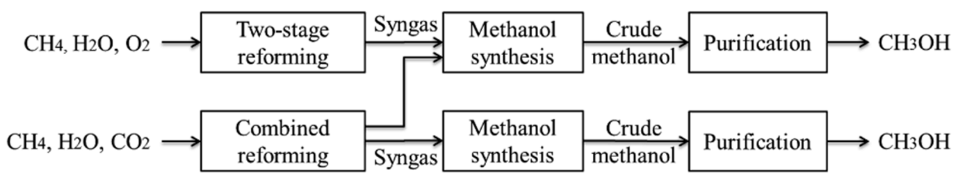

Figure 1. Schematic diagram of the integration of two methanol plants

4.2. Utilization of renewable energy sources for the CO_2 hydrogenation based methanol plant

Large demand of H_2, which is both expensive and laden with sizable CO_2 footprint, is the biggest hurdle to achieve a more sustainable solution for the methanol production via CO_2 hydrogenation. Instead of the conventional H_2 production via SMR, utilization of various renewable energy sources is an alternative worthy of consideration. Here, six different renewable energy sources for H_2 production are considered (Bartels et al., 2010; Yumurtaci and Bilgen, 2004): biomass, hydroelectric power, nuclear power, photovoltaic, solar thermal power, and wind power. For each case, the calculated net CO_2 flow and methanol production cost are plotted along with those of the SMR based H_2 production case and the conventional methanol plant case in Figure 2.

Most cases utilizing H_2 from renewable energy sources result in a reduction of net CO_2 flows except for the photovoltaic case. On the other hand, reasonable methanol production costs (less than 400 USD/t_{MeOH}) are obtained only in the case of hydroelectric power and biomass based H_2 production. The other cases are not found to be economically competitive because of their currently high power generation costs. Therefore, methanol production via CO_2 hydrogenation with H_2 production using hydroelectric power or biomass based is a more sustainable choice in terms of both the CO_2 flow and economic feasibility, at least at the current stage of development. Most likely, the wind and solar thermal power based H_2 production options can also become economically feasible if their power generation costs decrease further in the future.

5. Conclusion

CO_2 conversion technologies are currently attracting much interest due to the mounting concerns on climate change. To argue for the feasibility of CO_2 conversion processes, two promising reactions for methanol production, combined reforming and CO_2 hydrogenation, have been investigated. Plants employing the two reactions were designed and their net CO_2 flows and methanol production costs were evaluated and compared with those of the conventional methanol plant. The combined reforming based methanol plant showed an economic feasibility but it is found to release more CO_2 than it utilizes as input. In particular, direct CO_2 emission from the reformer furnace accounts for more than 90 % of the total CO_2 emission. For the case of CO_2 hydrogenation based methanol plant, the main problem was found to be the large demand for H_2 and its high cost and large CO_2 footprint. Since building a new installation of a standalone CO_2 conversion process based on these reactions is found to be not sustainable, two strategies to make them more sustainable have been investigated. First, integrating the combined reforming process with a methanol plant using the conventional two-stage reforming technology resulted in reduction of CO_2 emission as well as the methanol production cost saving. For the CO_2 hydrogenation based case, utilization of various renewable energy resources has shown to decrease the indirect CO_2 emission at the H_2 production step. In particular, use of the hydroelectric power or biomass in producing the required H_2 has led to a more sustainable solution. The interested reader can obtain the calculation details and simulation results by contacting the corresponding author.

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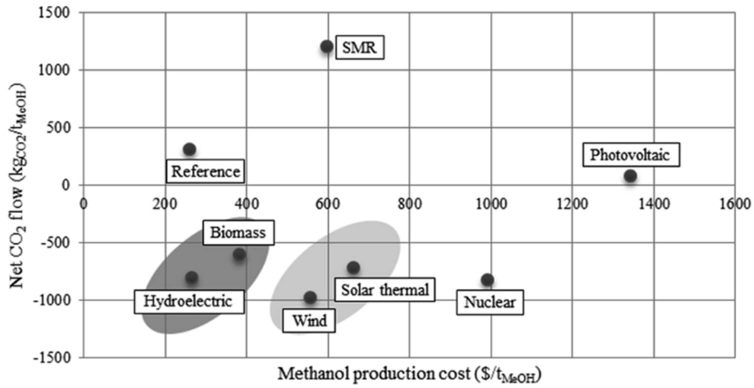

Figure 2. Net CO_2 flow and the methanol production cost of CO_2 hydrogenation based methanol plant with various H_2 production methods

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